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Course code: CSA0925

Course Name: Java programming

JAVA COLLECTIONS AND FRAMEWORK

Logger

LOGGER

You are working on a project that requires the use of a logging framework. Write a Java program to create an interface called "Logger" with methods for logging different types of messages, such as "debug", "info", "warning", and "error". Implement the interface in a class

```
1 import java.util.*;
2 interface Logger{
3     void debug(String msg);
4     void info(String msg);
5     void warning(String msg);
6     void error(String msg);
7 }
8 class Log4jLogger implements Logger{
9     ArrayList<String> logs=new ArrayList<>();
10    public void debug(String msg){
11        logs.add("[DEBUG]"+msg);
12    }
13    public void info(String msg){
14        logs.add("[INFO]"+msg);
15    }
16    public void warning(String msg){
17        logs.add("[WARNING]"+msg);
18    }
19    public void error(String msg){
20        logs.add("[ERROR]"+msg);
21    }
```

OUTPUT

```
[DEBUG]Program Started
[INFO]Data loaded
[WARNING]Battery low
[ERROR]Connnection failed timeout
```

Your INPUT goes here!

QUESTIONS 7 / 8 completed

HashSet

HASHSET

Write a Java program to create a HashSet of strings and remove a specific element from it.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         sc.nextLine();
7         HashSet<String> set = new HashSet<>();
8         for (int i = 0; i < n; i++) {
9             set.add(sc.nextLine());
10        }
11        String removeElement = sc.nextLine();
12        set.remove(removeElement);
13        for (String item : set) {
14            System.out.println(item);
15        }
16    }
17 }
```

INPUT

5
Apple
Banana
Mango
Orange
Grapes
Mango

OUTPUT

Apple
Grapes
Orange
Banana

QUESTIONS 8 / 8 completed

PriorityQueue

PRIORITYQUEUE

Write a Java program to create a PriorityQueue of integers and add elements to it.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         PriorityQueue<Integer> queue = new PriorityQueue<>();
7         for (int i = 0; i < n; i++) {
8             queue.add(sc.nextInt());
9         }
10        while (!queue.isEmpty()) {
11            System.out.print(queue.poll() + " ");
12        }
13    }
14 }
```

INPUT

5
30 10 50 20 40

OUTPUT

10 20 30 40 50

QUESTIONS 8 / 8 completed

TreeMap

TREEMAP

Write a Java program to create a TreeMap of strings and integers and iterate over its entries.

PROGRAM EDITOR

Java

Run

Save

```

1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         TreeMap<String, Integer> map = new TreeMap<>();
7         for (int i = 0; i < n; i++) {
8             String key = sc.next();
9             int value = sc.nextInt();
10            map.put(key, value);
11        }
12        for (Map.Entry<String, Integer> entry : map.entrySet()) {
13            System.out.println(entry.getKey() + " " + entry.getValue());
14        }
15    }
16 }

```

INPUT

```

3
Apple
10
Banana
20
Mango
30

```

OUTPUT

```

Apple 10
Banana 20
Mango 30

```

QUESTIONS 7 / 8 completed

Sort

SORT

Write a Java program to create a list of integers and sort them in ascending order using the Collections framework.

PROGRAM EDITOR

Java

Run

Save

```

1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         ArrayList<Integer> numbers = new ArrayList<>();
7         for (int i = 0; i < n; i++) {
8             numbers.add(sc.nextInt());
9         }
10        Collections.sort(numbers);
11        for (int num : numbers) {
12            System.out.print(num + " ");
13        }
14    }
15 }

```

INPUT

```

5
40 10 30 20 50

```

OUTPUT

```

10 20 30 40 50

```

Character

CHARACTER

You are developing a game that involves different types of characters, such as heroes, villains, and monsters. Write a Java program to create an interface called "Character" with methods for moving, attacking, and taking damage. Implement the interface in classes for each type of

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String heroName = sc.nextLine();
6         String villainName = sc.nextLine();
7         String monsterName = sc.nextLine();
8         Hero hero = new Hero(heroName);
9         Villain villain = new Villain(villainName);
10        Monster monster = new Monster(monsterName);
11        System.out.println("Hero:");
12        hero.move();
13        hero.attack();
14        hero.takeDamage();
15        System.out.println("Villain:");
16        villain.move();
17        villain.attack();
18        villain.takeDamage();
19
20        System.out.println("Monster:");
21        monster.move();
```

Superman
Joker
Godzilla

OUTPUT

```
Hero:
Superman moves forward.
Superman attacks the enemy.
Superman takes damage.
Villain:
Joker moves stealthily.
```

QUESTIONS 8 / 8 completed

LinkedListHashMap

LINKEDHASHMAP

Write a Java program to create a LinkedHashMap of strings and integers and retrieve the keys in the order they were inserted.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         LinkedHashMap<String, Integer> map = new LinkedHashMap<>();
7         for (int i = 0; i < n; i++) {
8             String key = sc.next();
9             int value = sc.nextInt();
10            map.put(key, value);
11        }
12        for (String key : map.keySet()) {
13            System.out.println(key);
14        }
15    }
16 }
```

INPUT

3
Apple
10
Banana
20
Mango
30

OUTPUT

```
Apple
Banana
Mango
```

QUESTIONS 8 / 8 completed

Max-Min

MAX-MIN

Write a Java program to create a Set of integers and perform different operations on it, such as finding the maximum and minimum values.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         Set<Integer> numbers = new HashSet<>();
7         for (int i = 0; i < n; i++) {
8             numbers.add(sc.nextInt());
9         }
10        int max = Collections.max(numbers);
11        int min = Collections.min(numbers);
12        System.out.println("Maximum: " + max);
13        System.out.println("Minimum: " + min);
14    }
15 }
```

OUTPUT

```
Maximum: 50
Minimum: 10
```

INPUT

```
5
10 40 20 50 30
```